

Student Visual Guide

A practical walkthrough of the AI Engineering Assessment Studio desktop application.

Assessment Studio

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Home

Session status, stage deadlines, connectivity, and alerts.

Workspace: C:\AI\Engineering\candidates\C014

The main workflow

Prepare the dataset, train and select a model, configure Vision and Calibration, practice with Robot Studio, then use the IDE and AI Assistant for candidate-owned work.

Start in Dataset and move forward only when the previous result is ready.

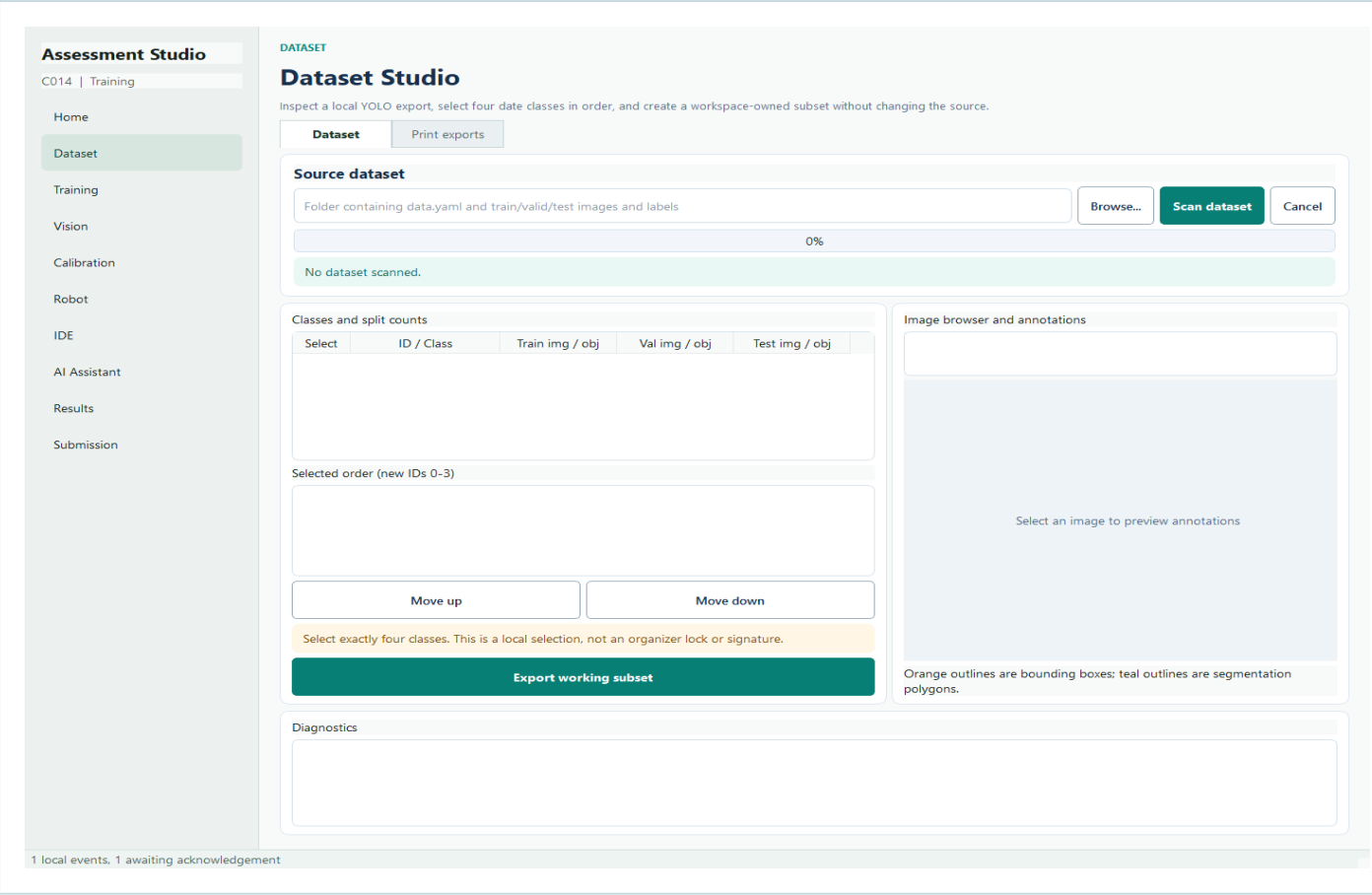
Windows launch

```
conda activate ai-assessment-studio
python -m ai_assessment.app
```

The teal buttons are primary actions, white outlined buttons are normal actions, red outlined buttons stop or caution, amber panels are hints, and green panels report status.

Dataset Studio

Inspect the source dataset, select exactly four classes, and export a workspace-owned subset.



How to use it

1. Enter the folder containing data.yaml and the train, validation, and test folders.
2. Press Scan dataset and resolve diagnostics before exporting.
3. Select exactly four classes and use Move up and Move down to set their order.
4. Review image overlays, then press Export working subset.
5. Use Print exports only when physical specimen cards are required.

Training Studio

Run recorded local YOLO experiments without downloading weights or freezing the interface.

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TRAINING

Training Studio

Record up to four distinct local YOLO experiments against the selected four-class subset. Validation results are candidate-visible only.

No complete four-class subset found under the candidate workspace.

Inputs

Subset

Base weights

Refresh subsets

Existing local .pt base weights; no downloads

Browse weights

Experiment configuration

Recorded slot

Epochs

Image size

Batch

Seed

Workers

Patience

Device

Experiment 1

10

640

8

0

0

20

blank = Ultralytics default, e.g. 0 or cpu

Start training

Cancel training

Refresh past runs

0%

Recorded runs

Run	Slot	Status	mAP50	mAP50-95
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Child-process log

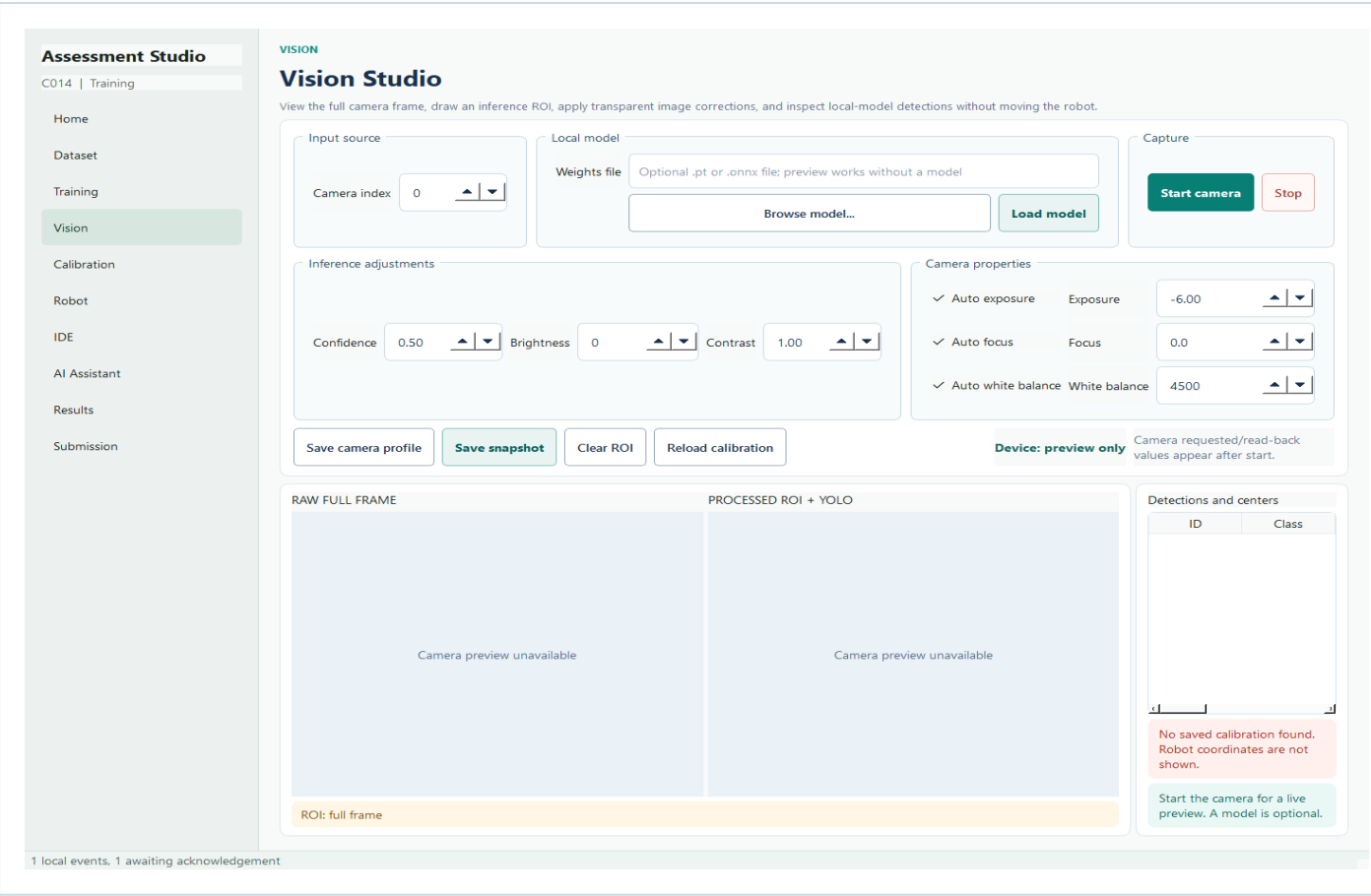
1 local events. 1 awaiting acknowledgement

How to use it

1. Choose the exported four-class subset and an existing local .pt base-weight file.
2. Configure one experiment slot, including seed, device, and epochs.
3. Press Start training and follow the live output.
4. Use Cancel training when needed; cancelled and failed runs remain recorded but are not completed experiments.
5. Repeat with four distinct configurations when the workflow requires four experiments.

Vision Studio

Preview the camera, draw the inference ROI, adjust processing, and inspect full-frame detections.



How to use it

1. Set Camera index, optionally load a local .pt or .onnx model, and press Start camera.
2. Drag on the raw image to draw an inference ROI; the processed image shows the YOLO input and annotations.
3. Adjust confidence, brightness, contrast, and supported camera properties.
4. Read full-frame boxes, centers, class IDs, confidence, and frame IDs in the detection table.
5. Save camera profile or snapshot when the setup is ready. Camera clicks never move the robot.

Calibration

Fit and verify the pixel-to-robot transform for the current camera geometry.

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VISION / ROBOTICS

Camera to robot calibration

Freeze a full-frame image, select four corresponding points, fit a transform, and test clicks without moving the robot.

01 Camera workspace

Camera index 0

Start camera

Freeze frame

Open saved image...

Start a camera or open an image
Then freeze the frame and select four points

ROBOT X: -- Y: --

Editing point 1: click its position in the image.

02 Calibrate the plane

Select a numbered row, click its marker in the image, and enter the matching robot X/Y. Hover the image for the 5x cursor magnifier.

POINT	PIXEL U	PIXEL V	ROBOT X	ROBOT Y
01	--	--	0.0	0.0
02	--	--	0.0	0.0
03	--	--	0.0	0.0
04	--	--	0.0	0.0

MATRIX TYPE

3 x 3 Perspective

2 x 3 Affine

Calculate matrix

Clear points

CAMERA PIXEL -> ROBOT PLANE

The matrix will appear here.

Displayed to 2 decimals; calculations and saved JSON retain full precision.

Save to workspace

Load workspace calibration

1 local events. 1 awaiting acknowledgement

How to use it

1. Start the camera or open a saved image, then freeze the frame.
2. Select four corresponding camera and robot points; use the magnifier for precise clicks.
3. Choose affine or homography and press Calculate matrix.
4. Click the image to test a predicted robot X/Y, then review the matrix and convention.
5. Press Save to workspace only after confirming the camera geometry is stable.

Robot Studio

Practice verified movement and supervised pick-and-place behavior in the local simulator.

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ROBOT

Robot Studio

Use the verified MaxArm HTTP contract for manual moves and supervised simulator trials.

Starting local simulator...

TARGET: LOCAL SIMULATOR

Connection and safety

TargetLocal simulatorBase URLhttp://127.0.0.1:51765Check healthDeep health

☐Expose simulator on LANEnable physical motionRead positions

Manual verified commands

LocationP1Suction on

HeighthighMoveSuction off

Duration1000 ms

Supervised simulator pick and place

ClassDestination

No exported classes foundRun supervised trial

SourceP1

DestinationP1Cancel next step

0%

Simulator-only XYZ

X-3 mm

Y-130 mm

Z89 mm

Move XYZ

Live simulator state

Current XYZ -

Target XYZ -

Motion progress -

Suction -

Objects -

Verified positions

	Position	Low X	Low Y	Low Z	High X	
1	P1	-3.0	-130.0	59.0	-3.0	-
2	P2	-124.0	-161.0	59.0	-124.0	-
3	P3	-124.0	-98.0	59.0	-124.0	-
4	P4	-124.0	-35.0	59.0	-124.0	-
5	P5	-124.0	28.0	59.0	-124.0	2

Simulator command history

1 local events. 1 awaiting acknowledgement

How to use it

1. Start with Local simulator, then press Check health and Read positions.
2. Use high poses for travel and low poses only at a source or destination.
3. Use suction controls deliberately and map each class to a destination.
4. Run supervised trial to exercise the complete clearance route.
5. Physical MaxArm mode starts disarmed and requires explicit operator enabling.

IDE And Terminal

Edit, save, run, and inspect candidate-owned Python code inside the workspace.

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IDE

Edit and run candidate-owned Python files from workspace/src. Runs are local process executions, not an OS security sandbox.

Candidate workspace

Name	Size	Type
> calib...		File Folc
> cam...		File Folc
> data...		File Folc
> expo...		File Folc
> logs		File Folc
> mod...		File Folc
> results		File Folc
> src		File Folc
> sub...		File Folc
proj...	235 bytes	applicat
sessi...	246 bytes	applicat

New Python file

Python editor

Candidate files under src/

Save Save As Run Python Stop Attach selection Ask AI

Arguments: optional arguments, parsed int...

Run output

No Python run active

Python stdout and stderr will appear here.

Terminal

pipe-based process Stopped

Start Restart Stop

This terminal uses pipes: interactive console control sequences and secure prompt behavior may be limited. Input is never recorded.

command Send

1 local events. 1 awaiting acknowledgement

How to use it

1. Double-click a file in the candidate workspace tree or press New Python file.
2. Use Save or Save As; paths outside the workspace are rejected.
3. Press Run Python for the saved file and watch stdout, stderr, and tracebacks in Run output.
4. Use Stop only for the IDE process. It does not stop Training.
5. Start the genuine pipe-based Terminal when interactive commands are needed.

AI Assistant

Ask the organizer gateway for help using context that you explicitly review and send.

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AI Assistant

Organizer gateway only. Attach context deliberately; the desktop never holds an OpenAI key and is not a security boundary.

Gateway URL

http://127.0.0.1:8765

Session credential

practice session credential from the gateway operator

Connect / refresh

Not connected

Model: unavailable Allowance: unavailable

Conversation history

Generated code blocks

New request

Ask a question for the approved organizer model...

Attached context:

Selected IDE text

Use IDE selection

Nothing will be sent until you add context here. The preview is exactly what will be attached.

Send

Ready

1 local events. 1 awaiting acknowledgement

How to use it

1. Enter the gateway URL and session credential, then press Connect / refresh.
2. Choose a context type, inspect the exact preview, and write a question.
3. Press Send deliberately; the client never calls OpenAI directly.
4. Use Send to IDE on a specific fenced code block in the response.
5. Choose New file, Insert at cursor, or Replace selection. Generated code stays unsaved and is never run automatically.

Results Studio

Compare candidate-visible validation results and select the production checkpoint.

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RESULTS

Results Studio

Compare recorded validation runs and select one exact workspace checkpoint for Vision. These are candidate-visible results; organizer scores and locks are not present here.

No recorded training runs yet.

Refresh results

Select production model

Candidate-visible validation comparison

Run	Status	mAP50	mAP50-95	Precision	Recall	Latency ms	Checkpoints	Selected
-----	--------	-------	----------	-----------	--------	------------	-------------	----------

Selected run details

Select a completed run.

Class	mAP50	mAP50-95	Precision	Recall
-------	-------	----------	-----------	--------

Artifacts

Confusion matrix: unavailable

1 local events. 1 awaiting acknowledgement

How to use it

1. Review recorded runs and the metrics that are actually available.
2. Treat unavailable values as unavailable; do not infer hidden scores.
3. Select a completed checkpoint with Select production model.
4. The selection records the run identity and exact checkpoint hash for later Vision use.

Submission

Understand the boundary of the current local application.

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Submission

Validate and lock the final evidence snapshot.

Workspace: C:\AIEngineering\candidates\C014

1 local events. 1 awaiting acknowledgement

How to use it

1. The Submission page is reserved for the later organizer-controlled assessment flow.
2. Local events, candidate-visible metrics, and simulator results are not a server receipt.
3. Follow the organizer instructions when the official assessment integration is enabled.

Safety And Working Rules

Keep these boundaries in mind while practicing:

- The original dataset is read-only. Exported subsets and generated artifacts belong under the candidate workspace.
- The AI gateway owns the provider key. Never put a provider key in the desktop application or candidate scripts.
- Vision and calibration never send robot commands automatically.
- Physical robot motion remains supervised and starts disarmed. Keep the physical emergency stop available.
- Candidate code runs locally and is not an operating-system security sandbox.
- The local event log is evidence capture, not an organizer server receipt.